LCS2
LABORATORY FOR COMPUTATIONAL SOCIAL SYSTEMS

Explainability

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Dictionary meaning of “Explanation”

- **English wordnet** - explanation, account (a *statement* that makes something comprehensible by describing the relevant *structure* or *operation* or *circumstances* etc.)
  - "the explanation was very simple";
  - "I expected a brief account"
- **Oxford Learner Dictionary** - a statement, fact, or situation that tells you *why something happened*; a *reason given for something*:
  - “The most likely explanation is that his plane was delayed”.
- **Longman Dictionary** - The reasons you give for *why something happened* or *why you did something*.

Structure

- Explanation- Keep reducing to simpler and simpler statements, until self evident
    - <something> happened due to *delayed delivery*
    - Delay happened due to airport *congestion*
    - Congestion happened due to *election rallies*
    - Election rallies happened due to the need for *attracting votes*
    - Attracting votes happened due to *desire for winning election*
    - ....

Operation

- Explanation- describes “how” or “How something happened”
  - How did you fix the deep fridge?!”
    - Switched off power
    - Let the ice melt
    - Switched power back on
    - …

Circumstances

- Explanation- describes “why”
  - What circumstances led to the current circumstance?
  - Overlap with “structure based”
  - Covid-19 resurgence- circumstance of interest
    - Opening of facilities
    - Casual behaviour

Not necessary a chain always, many factors work together!

Explainability in ML

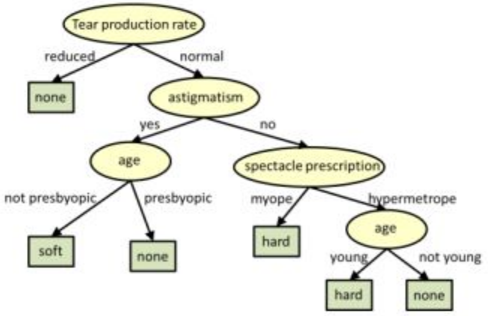
- Primarily, SENSITIVE ANALYSIS to input features.
  - Establishes Correlation of decision with input parts and features
  - Not causality analysis

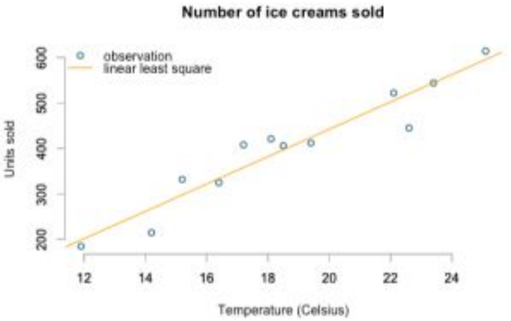
A scientist would want deeper insight- not just “co-occurrence”

Explainability in ML

Interpretable Models

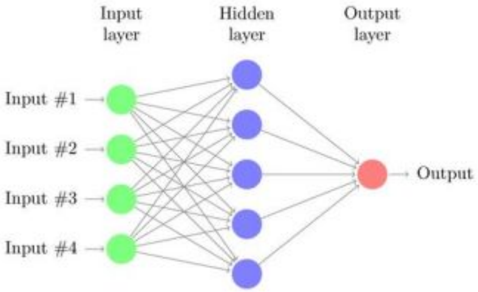
High Explainability, Low Accuracy, Low Robustness

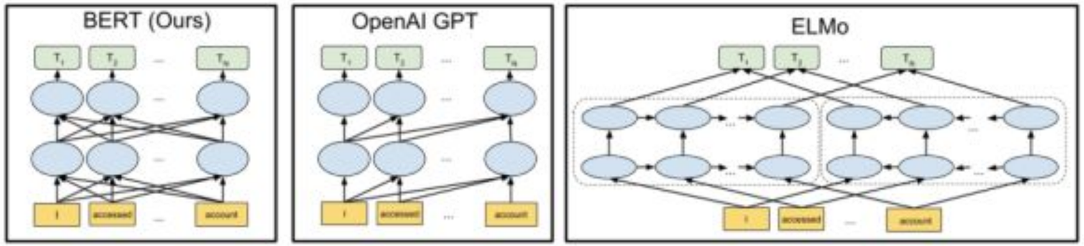




Black boxes

Low Explainability, High Accuracy, High Robustness





Black box Mishaps

- Prevents widespread adoption of black boxes in mission-critical areas
- Even in other application areas, such mistakes can make/break companies

https://www.bbc.com/news/technology-33347866

https://edition.cnn.com/2016/12/07/asia/new-zealand-passport-robot-asian-trnd/index.html

Black box Mishaps in NLP

- Emotion analysis on the input
  - *The conversation with Justin was exciting.* ⇒ Joy
  - *The conversation with Malik was exciting.* ⇒ Anger

Example 1: True Label = Joy, Pred Label = Joy						
Sentence	The	conversation	with	Justin	was	exciting
Expected Focus	0	0	0	0	0	1
Actual Focus	0	0	0	0.9	0	0.1

Example 2: True Label = Joy, Pred Label = Anger						
Sentence	The	conversation	with	Malik	was	exciting
Expected Focus	0	0	0	0	0	1
Actual Focus	0	0	0	0.8	0	0.2

- Kiritchenko and Mohammad (2018)
  - Evaluated 215 sentiment/emotion analysis systems
  - Equity Evaluation Corpus (EEC)
  - 200 of these suffered from such biases.

Motivation for Explainability

- Increase Trust
- Reveal Causality
  - ML models- mainly learn correlation, and not cause
- Identify and mitigate biases
- Help decide necessary and sufficient set of features for the decision making task
  - Uncover feature importance
- Extract knowledge (Lipton et al., 2016)

Closely-related terms under the realm of Explainability

- **Logic and Philosophy**
  - Axioms
  - Self evident
  - Inference
  - Reasoning
  - Correlation
  - Coincidence
  - Causality
- **AI-ML**
  - Trustworthiness
  - Fairness
  - Bias
  - Ethics
  - Interpretability
  - Transparency
  - Outlier
  - Anomaly

Some existing and popular explainability models/tools

- Heatmap Analysis of Attention mechanism
- LIME (*Locally Interpretable Model Agnostic Explanations*) [Riberio et al.,2016, KDD]
    - LIME generates additional samples (fake data) by perturbation, and uses the model to classify them.
    - Returns the weights of this classifier as feature importance explanation for the particular datapoint
- LRP (*Layer-wise relevance propagation*) [Bach et al., PloS One]
    - Explains by redistributing the final prediction output back in the network
    - Input neurons that contribute the most to higher layer get max relevance
- SHAP (*SHapley Additive exPlanations*) [Lundberg and Lee, 2017, NIPS]
    - Assigns each feature an importance value for a particular prediction
- ..

https://github.com/slundberg/shap

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Nice perfume. How long did you marinate in it? Multimodal Sarcasm Explanation

AAAI-2022

Poorav Desai, Tanmoy Chakraborty, Md Shad Akhtar*

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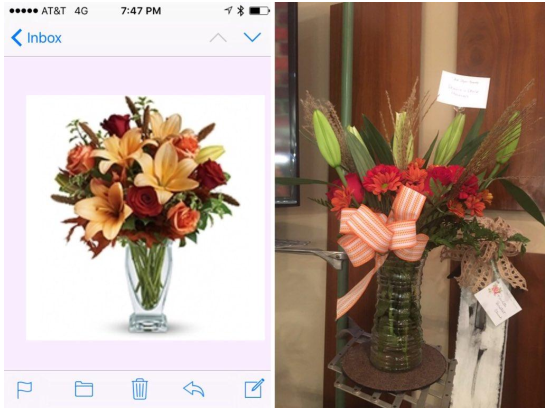
Multimodal Sarcasm

- Sarcasm: Satirical or ironic statements usually to *hurt*, *insult*, or *offend* someone.
  - *Incongruity exists between the surface meaning and the intended meaning*
- Multimodal Sarcasm: Two or more modalities (viz. *image*, *text*, *audio*, etc.) work together
  - *Inter- and intra-modality incongruity.*

This guy gets a gold star for excellent parking.

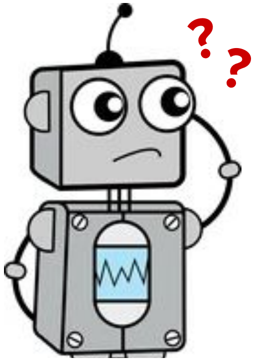


Nailed it! florist thinks this is acceptable <user> #nailedit #greatservice

A crucial observation

- Sarcasm is highly contextual.
- Despite excellent performance of detection systems, at times, it is difficult to comprehend why something is sarcastic.

In real-world, we ask our friends or family to explain it.



Why can't a system do the same?

Multimodal Sarcasm Explanation (MuSE)

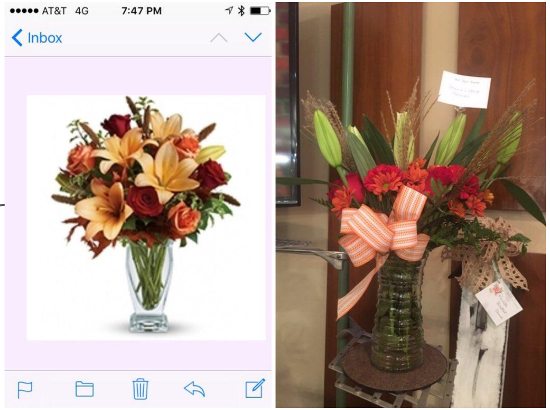
- Sometime it is challenging to comprehend the underlying irony
    - Complex nature of sarcasm
    - Lack of adequate context
- Explanation will help understanding the sarcasm.

This guy gets a gold star for excellent parking.

This guy has parked his car partially covering the parking slot for handicapped.

Explanations

This isn't acceptable since the product advertised isn't what has been delivered.



Nailed it! florist thinks this is acceptable <user> #nailedit #greatservice

MuSE: Problem Formulation

- For a given multimodal sarcastic post $P = <I, T[t_1, t_2, ..., t_N]>$, where I and $T[]$ are image and caption, respectively, and t_i is the token in the caption, MuSE aims to generate a natural language explanation $E[e_1, e_2, ..., e_D]$ revealing the intended irony, where $\forall\, t_i, e_j \in \mathrm{Vocab}^{English}$.

Image I

Caption T This guy gets a gold star for excellent parking.

Explanation E

This guy has parked his car partially covering the parking slot for handicapped.

Novelty of MuSE and Related works

- Sarcasm Understanding:
  - Sarcastic to Non-Sarcastic Interpretations (Dubey, Joshi, and Bhattacharyya, 2019; Peled and Reichart, 2017)
    - Converts a given *sarcastic* text into its *non-sarcastic* interpretation ⇒ *Negates the text*
  - Sarcasm Generation (Mishra, Tater, and Sankaranarayanan, 2019):
    - Generates *sarcastic* text given a *negative sentiment* sentence
- Explainability:
  - Majority of the explainability research to explain the model behaviour revolve around ***attention heatmaps*** (Guo et al. 2019; Yao and Wan, 2020) or similar mechanisms, e.g., *LIME* (Pramanick et al. 2021; Mahajan, Shah, and Jafar, 2021), *SHAP* (Parsa et al. 2020), etc.

MuSE:

- Natural language generation for explanation (explanations are more than just non-sarcastic interpretation)
- Multimodality

*https://github.com/LCS2-IIITD/Multimodal-Sarcasm-Explanation-MuSE

MORE* Dataset

- **Curated** 22K sarcastic posts from existing Multimodal Sarcasm Detection datasets
  - Schifanella et al., 2016; Sangwan et al., 2020; and Headacheboy[1]
- **Excluded** ~18K posts due to one of the following issues:
  - Non-sarcastic posts are discarded.
  - *Posts with explicit mention of sarcasm are discarded.*
  - Posts with non-English content are discarded.
  - Posts that need additional context to interpret sarcasm or the annotators are not familiar with the topics are discarded.
- **Generated explanations** for the remaining 3510 posts as per the following guidelines.
  - All entities including image, caption, hashtags, emojis, etc., are to be considered for interpreting the irony and generating an appropriate explanation.
  - In case the underlying sarcasm can be explained in multiple ways, the shorter and simpler explanation is preferred.
  - Any unrelated topic in explanation is avoided.

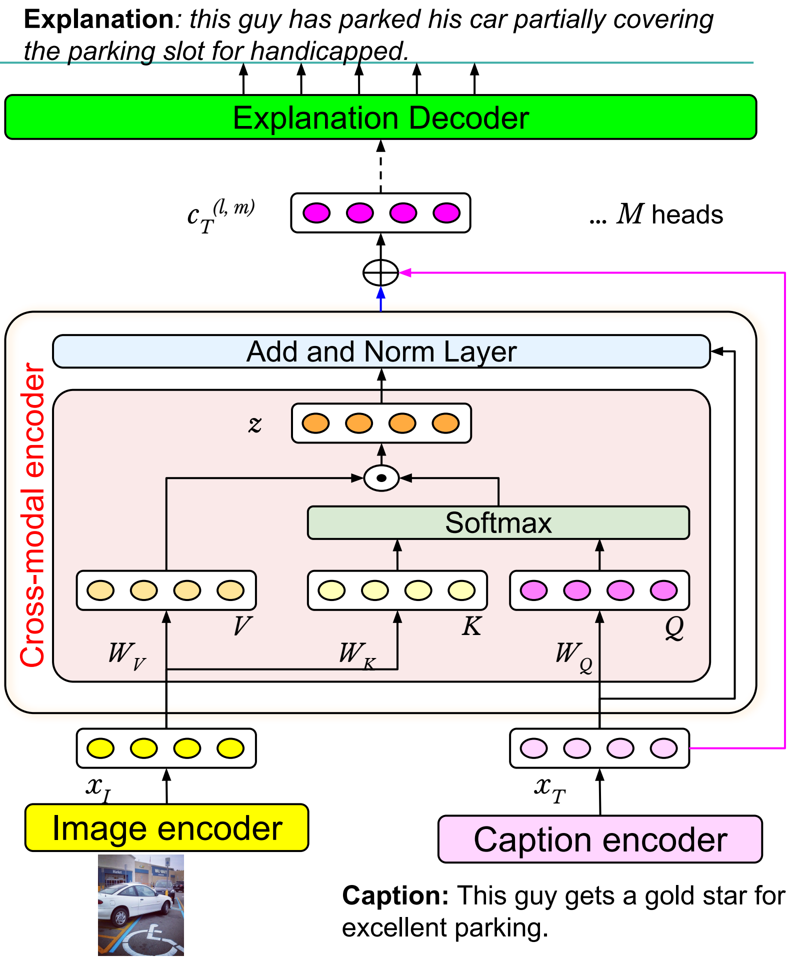
Some idiots just don't get it

Split	# of Posts	Caption		Explanation	
		Avg. Len	$\|V\|$	Avg. Len	$\|V\|$
Train	2983	19.75	9677	15.47	5972
Val	175	18.85	1230	15.39	922
Test	352	19.43	2172	15.08	1527
Total	3510	19.68	10865	15.43	6669

[1] https://github.com/headacheboy/data-of-multimodal-sarcasm-detection

Proposed Benchmark: ExMORE

- Image encoder: Pretrained VGG-19
- Caption encoder: Pretrained BART-base encoder
- Cross-modal encoder
- Explanation decoder: Pretrained BART-base decoder



Experimental Setup

- Three setups
  - Only OCR samples (1968), Only non-OCR (1542) samples, and the complete dataset (3510)
- Evaluation metrics
      - BLEU (B1, B2, B3, B4),
      - ROUGE (R1, R2, RL),
      - METEOR,
      - BERTScore (Zhang et al., 2020), and
      - SentBERT (Cosine)
- Comparative systems – Adapted for MuSE
  - Unimodal: Pointer Generator Network (See, Liu, and Manning 2017), Transformer (Vaswani et al. 2017)
  - Multimodal: MMFG video summarisation systems (Liu et al., 2020), Multimodal Transformer (M-transf)

Experimental Results

Model	BLEU				Rouge			METEOR	BERT-Score			Sent-BERT
	B1	B2	B3	B4	R1	R2	RL		Pre	Rec	F1	(Cosine)
Pointer Generator Network	17.54†	6.31	2.33	1.67†	17.35	6.90	16.00	15.06†	84.8	85.1	84.9	49.42
Transformer	11.44	4.79	1.68	0.73	17.78	5.83	15.90	9.74	83.4	84.9	84.1	52.55
MFFG-RNN	14.16	6.10	2.31	1.12	17.47	5.53	16.21	12.31	81.5	84	82.7	44.65
MFFG-Transf	13.55	4.95	2.00	0.76	16.84	4.30	15.14	10.97	81.1	83.8	82.4	41.58
M-Transf	14.37	6.48†	2.94†	1.57	20.99†	6.98†	18.77†	12.84	86.3†	86.2†	86.2†	53.85†
ExMore	**19.26**	**11.21**	**6.56**	**4.26**	**27.55**	**12.49**	**25.23**	**19.16**	**88.3**	**87.5**	**87.9**	**59.12**

(a) All samples

Model	BLEU				Rouge			METEOR	BERT-Score			Sent-BERT
	B1	B2	B3	B4	R1	R2	RL		Pre	Rec	F1	(Cosine)
Pointer Generator Network	17.87†	6.37	1.92	1.36†	17.80	6.92	16.43	15.62†	84.7	85.2	84.9	48.77
Transformer	11.65	5.65	1.73	0.69	17.41	6.26	16.16	10.13	83.6	85.1	84.3	48.40
MFFG-RNN	15.43	6.82	2.46	1.33	18.61	5.71	17.40	12.98	81.6	84.3	82.9	42.72
MFFG-Transf	13.28	5.35	1.49	0.26	16.80	4.35	14.90	11.19	81.3	84	82.6	41.68
M-Transf	14.91	6.90†	2.66†	0.83	21.05†	7.08†	19.34†	13.91	86.5†	86.3†	86.4†	51.77†
ExMore	**19.47**	**11.69**	**6.82**	**4.27**	**27.12**	**12.12**	**24.92**	**19.20**	**88.3**	**87.6**	**88.0**	**56.95**

(b) Non-OCR samples

Model	BLEU				Rouge			METEOR	BERT-Score			Sent-BERT
	B1	B2	B3	B4	R1	R2	RL		Pre	Rec	F1	(Cosine)
Pointer Generator Network	17.19†	6.08	2.49	1.79	16.92	6.76	15.55	14.64†	84.9	84.9	84.9	49.53
Transformer	10.68	4.01	1.49	0.71	17.25	5.32	15.04	8.99	83.2	84.7	83.9	53.94
MFFG-RNN	12.18	4.92	1.73	0.88	15.18	4.56	14.01	10.64	81.2	83.7	82.4	45.91
MFFG-Transf	12.87	4.12	1.69	0.62	15.54	3.53	14.20	9.70	81	83.6	82.3	41.13
M-Transf	14.06	6.25†	3.22†	2.28†	21.04†	7.01†	18.42†	12.06	86.2†	86.1†	86.1†	55.66†
ExMore	**19.40**	**11.31**	**6.83**	**4.76**	**28.02**	**13.10**	**25.66**	**19.55**	**88.2**	**87.5**	**87.9**	**60.82**

(c) OCR samples

Ablation Study

- We tried one more variant by including image description as an additional modality
  - Microsoft OSCAR for the image description
- Though some of the descriptions were relevant, overall, it lacked relevant features contributing to irony
  - Performance was not good.

Caption: This guy gets a gold star for excellent parking.

Desc: A white car parked in a parking lot.

Result Analysis - Linguistic view

- POS tag based difference and overlap counts

Model	Total		Non-OCR		OCR	
	M-Transf	ExMore	M-Transf	ExMore	M-Transf	ExMore
Ref count	3.76		3.75		3.80	
Gen count	3.68	3.57	3.62	3.53	3.73	3.68
Difference	1.80	1.81	1.80	1.88	1.80	1.76
Overlap	0.68	1.03	0.63	1.05	0.73	1.06
Overlap–Syn	0.79	1.18	0.73	1.19	0.85	1.21

(a) Noun

Model	Total		Non-OCR		OCR	
	M-Transf	ExMore	M-Transf	ExMore	M-Transf	ExMore
Ref count	2.78		2.71		2.80	
Gen count	2.67	2.41	2.51	2.38	2.81	2.46
Difference	1.24	1.18	1.16	1.15	1.32	1.17
Overlap	0.45	0.60	0.38	0.51	0.52	0.69
Overlap–Syn	0.60	0.77	0.52	0.72	0.67	0.81

(b) Verb

Model	Total		Non-OCR		OCR	
	M-Transf	ExMore	M-Transf	ExMore	M-Transf	ExMore
Ref count	1.32		1.19		1.40	
Gen count	0.86	1.04	0.80	0.91	0.91	1.14
Difference	1.04	0.91	0.91	0.85	1.13	0.95
Overlap	0.04	0.16	0.05	0.15	0.04	0.17
Overlap–Syn	0.04	0.18	0.05	0.16	0.04	0.20

(c) Adjectives

Model	Total		Non-OCR		OCR	
	M-Transf	ExMore	M-Transf	ExMore	M-Transf	ExMore
Ref count	1.03		0.96		1.09	
Gen count	0.78	0.69	0.73	0.61	0.80	0.76
Difference	0.79	0.72	0.84	0.71	0.76	0.74
Overlap	0.27	0.24	0.22	0.18	0.31	0.28
Overlap–Syn	0.27	0.24	0.22	0.18	0.31	0.28

(d) Adverb

- Significant gap b/w the generated and reference explanations, but ExMore is better than M-transf in all cases.
- Improvements in Overlap-Syn count suggest ExMore's explanations are slightly better at the semantic level as well.

Result Analysis - Human Evaluation

- Randomly sampled 50 generated explanations and asked 25 evaluators to rate them
  - **<u>Adequacy:</u>** Goodness of explanation to reveal the underlying sarcasm.
    - Justify [1.0], Weakly Justify [0.66], Somewhat Related to Input (SRI) [0.33], and Not Related to Input (NRI) [0.0]
  - **<u>Fluency:</u>** Coherency of English explanation.
    - Fluency is rated on the continuous scale of [0, 1].

Model	Adequacy	Fluency
M-Transf	0.37	0.74
ExMore	0.69	0.89

(a) Adequacy and Fluency scores.

Model	Adequacy Rating			
	Justify	W. Justify	SRI	NRI
M-Transf	15%	15%	35%	35%
ExMore	65%	5%	20%	10%

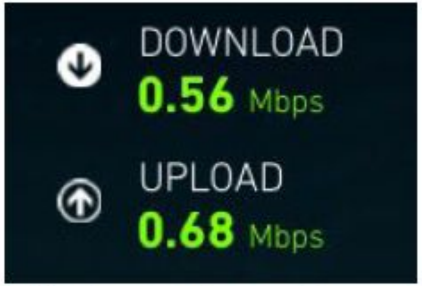
(b) Adequacy rating distribution of M-transf and ExMore.

Model	Adequacy	Fluency
M-Transf	0.367	0.221
ExMore	0.401	0.175

(c) Fleiss' Kappa scores among 25 human evaluators.

*Majority-voting

Result Analysis - Human Evaluation



Caption: dear <user> please never change . you are perfect in every way .

Ground truth: the author isn't happy with <user>'s internet speeds.

Explanation: the author is pissed at <user> for such terrible internet speed.

(a) Justify

Caption: really enjoying the # atmfcb game tonight <user> <user>

Ground truth: the author can't enjoy the atmfcb game tonight because of such disturbance.

Explanation: the author is pissed at <user> for such disturbance.

(b) Weakly Justify

Caption: i just love paying for chunks of ice ... i mean a smoothie <user> # metrocenter # unacceptable

Ground truth: the author hates paying for chunks of ice in the name of a smoothie.

Explanation: the author is pissed at <user> for having to wait for a few pieces of ice.

(c) SRI

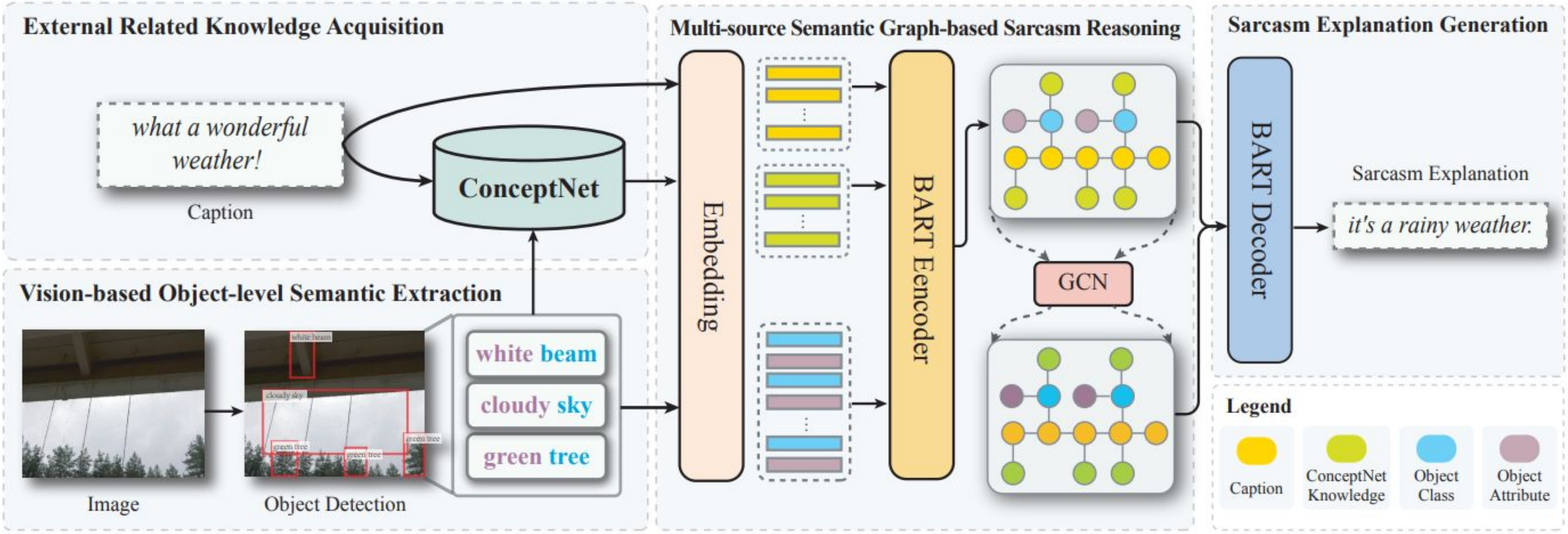
Caption: i think i may be the greatest fisherman who has ever lived. just look at the size of my largemouth bass.

Ground truth: the size of the author's largemouth bass is very small.

Explanation: it's obvious from the stats.

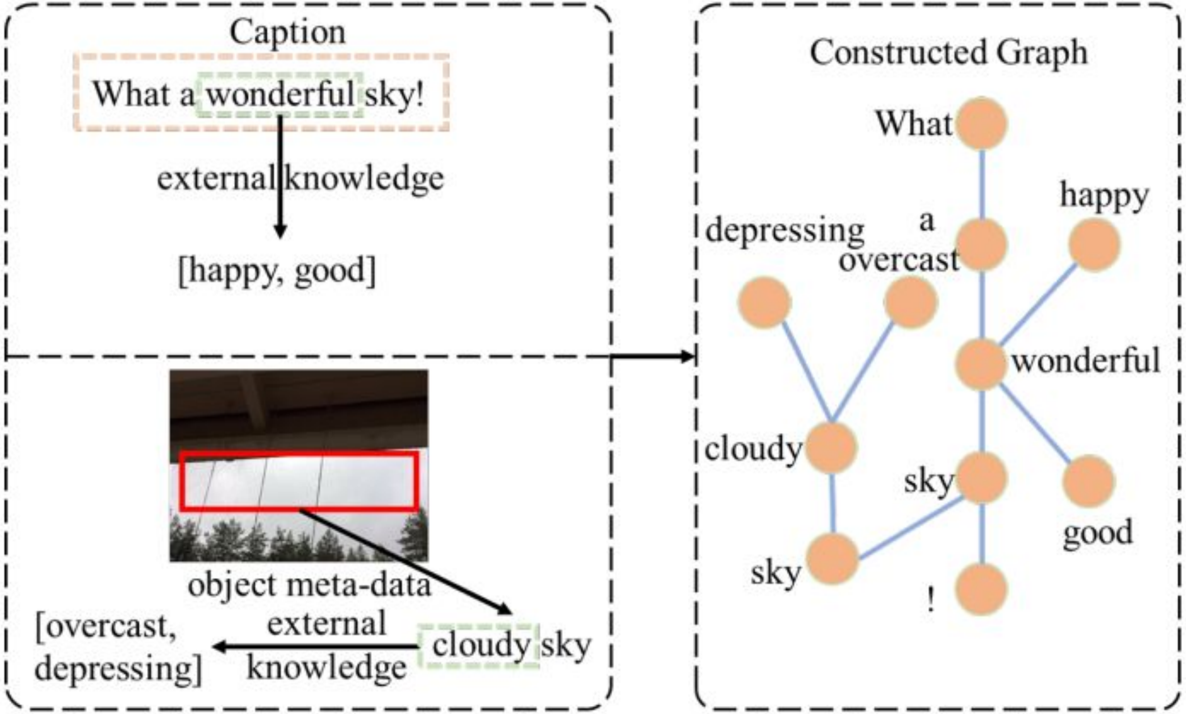
(d) NRI

Multi-source Semantic Graph-based Multimodal Sarcasm Explanation Generation, ACL-2023



Multi-source Semantic Graph-based Multimodal Sarcasm Explanation Generation, ACL-2023

- Multi-source semantic graph construction process



Multi-source Semantic Graph-based Multimodal Sarcasm Explanation Generation, ACL-2023

Model	BLEU				Rouge			METEOR	BERT-Score			Sent-BERT
	B1	B2	B3	B4	RL	R1	R2		Pre	Rec	F1	(Cosine)
PGN	17.54	6.31	2.33	1.67	16.00	17.35	6.90	15.06	84.80	85.10	84.90	49.42
Transformer	11.44	4.79	1.68	0.73	15.90	17.78	5.83	9.74	83.40	84.90	84.10	52.55
MFFG-RNN	14.16	6.10	2.31	1.12	16.21	17.47	5.53	12.31	81.50	84.00	82.70	44.65
MFFG-Transf	13.55	4.95	2.00	0.76	15.14	16.84	4.30	10.97	81.10	83.80	82.40	41.58
M-Transf	14.37	6.48	2.94	1.57	18.77	20.99	6.98	12.84	86.30	86.20	86.20	53.85
ExMore	19.26	11.21	6.56	4.26	25.23	27.55	12.49	19.16	88.30	87.50	87.90	59.12
TEAM-w/o-Know	52.63	42.42	35.80	30.91	48.67	49.28	33.18	48.53	90.90	91.40	91.10	71.58
TEAM	**55.32**	**45.12**	**38.27**	**33.16**	**50.58**	**51.72**	**34.96**	**50.95**	**91.80**	**91.60**	**91.70**	**72.92**

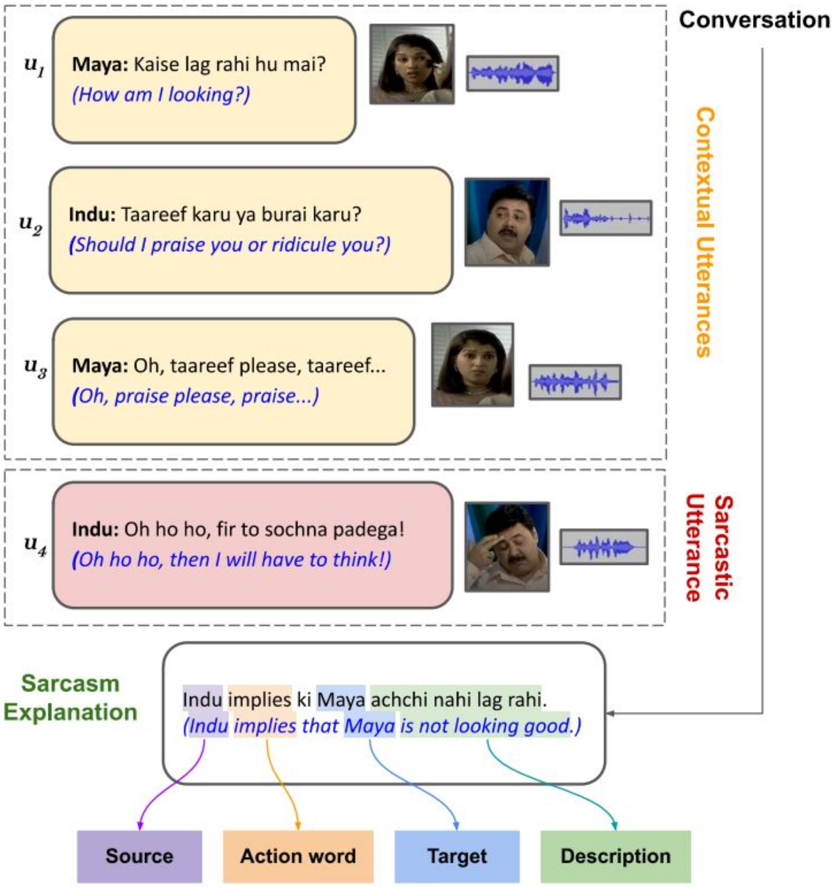
Sarcasm Explanation in Dialogs (SED)

ACL-2022

With Shivani, Atharva, and Tanmoy

Sarcasm Explanation in Dialogues (SED)

- Given a multi-modal code-mixed sarcastic dialogue, produce a natural language explanation for the sarcasm present in it



Literature

- Only sarcasm detection [Joshi et al. 2017; Chauhan et al. 2020]
- With unimodal data [Bharti et al. 2017; Swami et al. 2018]
- Sarcasm generation [Mishra et al. 2019]

✓ Sarcasm explanation

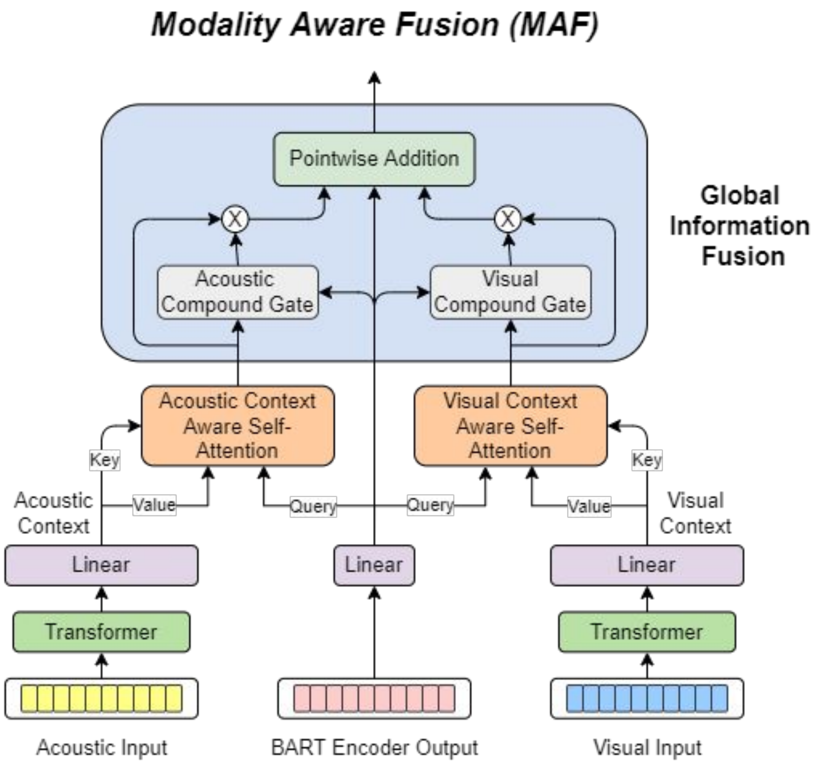
✓ Hindi-English code-mixed, multi-modal input

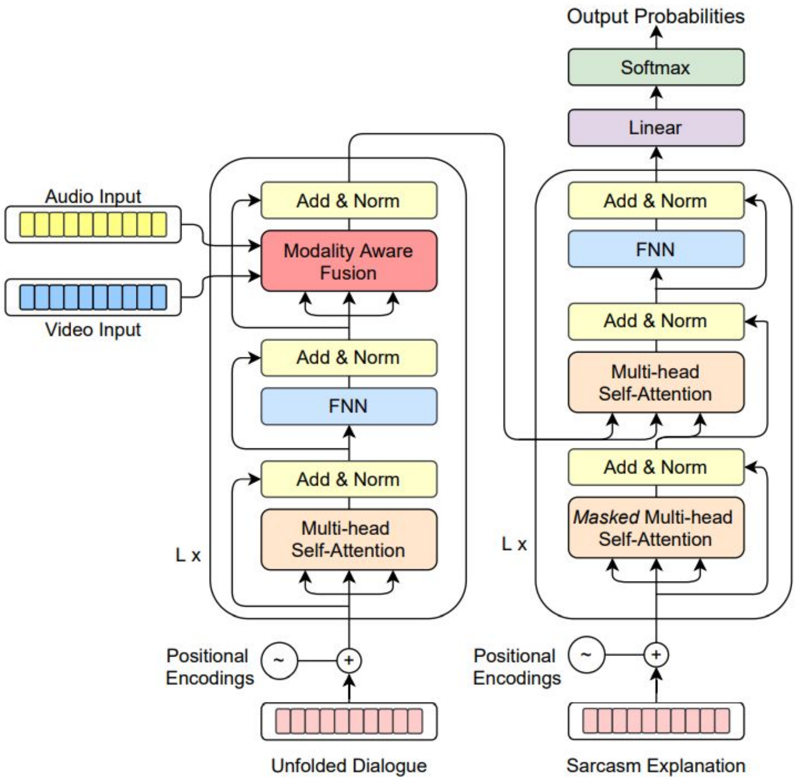
Dataset- WITS

- Extension of MaSaC dataset
- Collected better resolution videos
- Cleaned → sarcastic statements selected → context defined → annotated with explanation

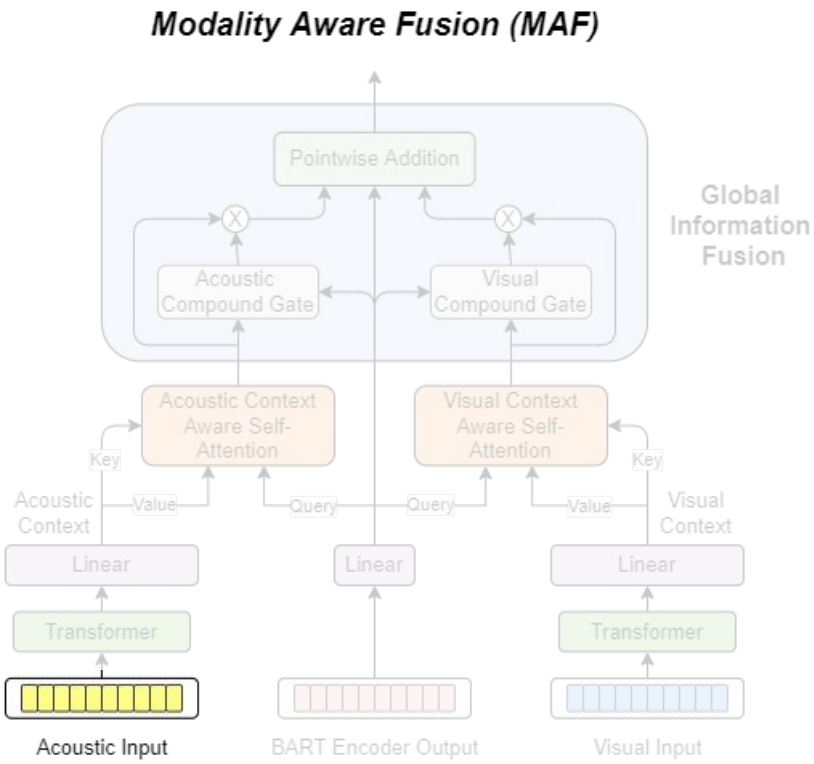
# Dlgs	**# Utts**	**# Eng utts**	**# Hin utts**
2240	9080	101	1453
# CM utts	**Avg. utt/dlg**	**Avg. sp/dlg**	**Avg. words/utt**
7526	4.05	2.35	14.39
Avg. words/dlg	**Vocab size**	**Eng vocab size**	**Hin vocab size**
58.33	10380	2477	7903

Proposed Architecture: MAF-TAV



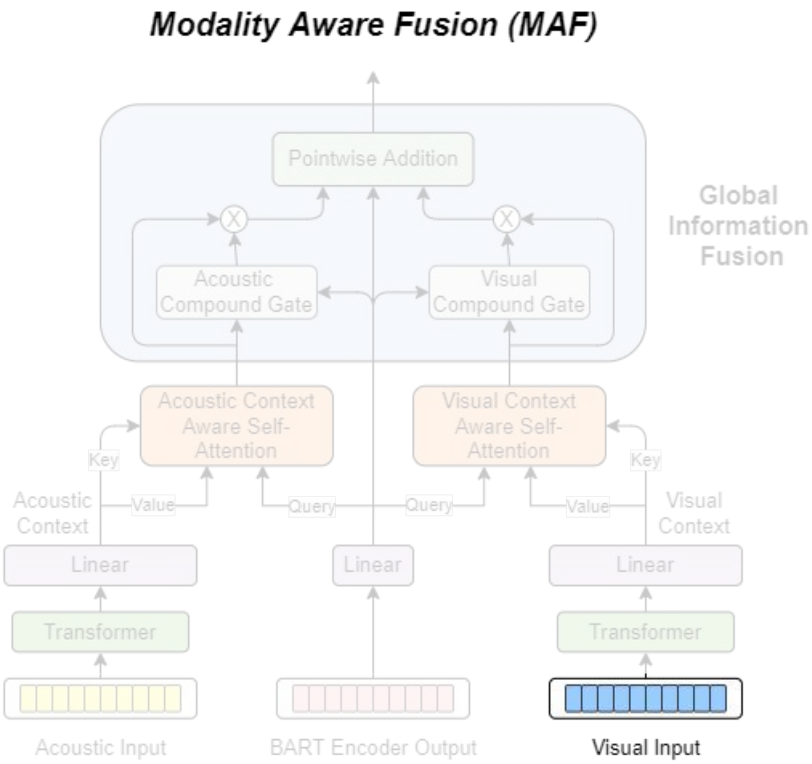


Proposed Architecture: MAF-TAV



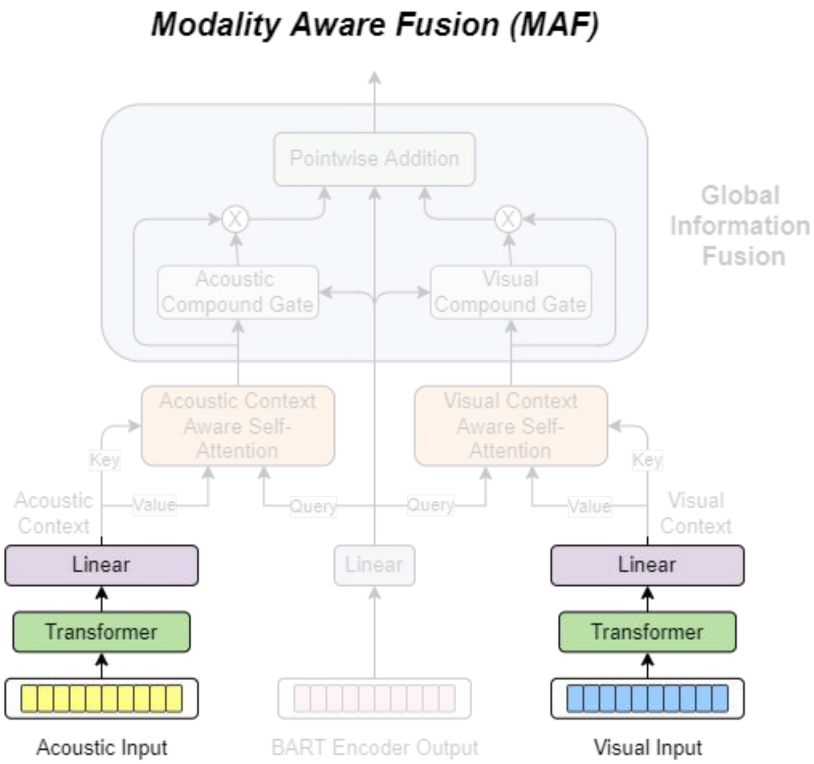
- MFCCs from openSMILE

Proposed Architecture: MAF-TAV

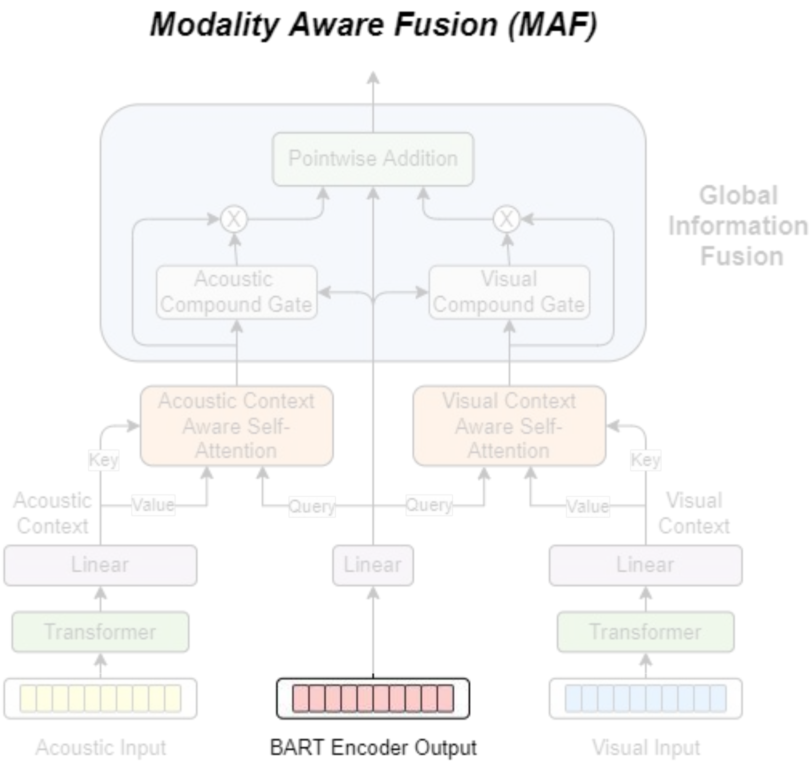


- Pretrained ResNet

Proposed Architecture: MAF-TAV

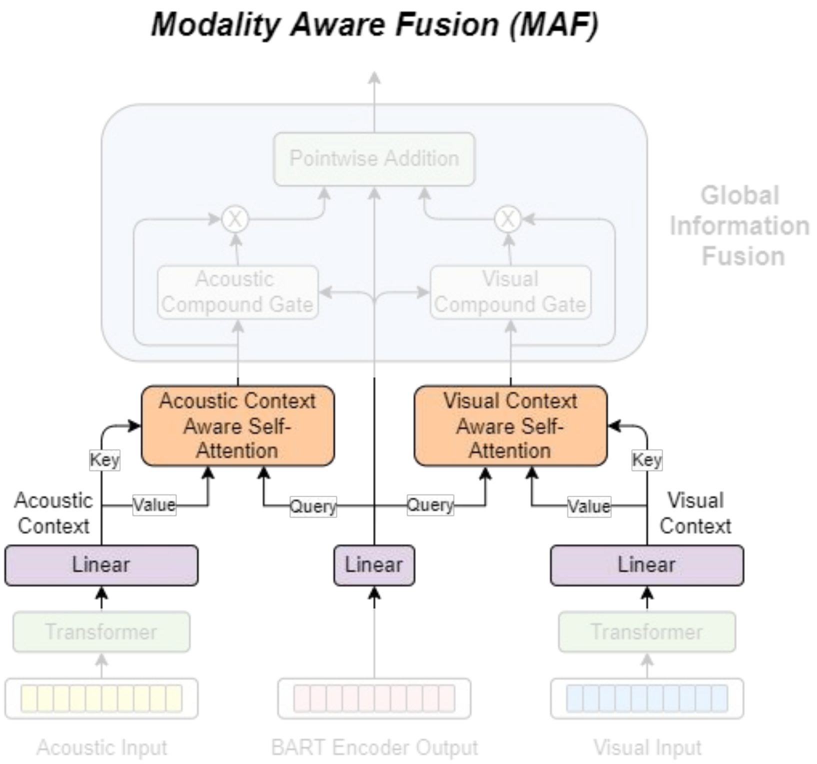


Proposed Architecture: MAF-TAV



- BART encoder

Proposed Architecture: MAF-TAV



- Calculating key and value using multimodal information

$$\begin{bmatrix}\hat{K}\\ \hat{V}\end{bmatrix} = (1 - \begin{bmatrix}\lambda_k\\ \lambda_v\end{bmatrix}) \begin{bmatrix}K\\ V\end{bmatrix} + \begin{bmatrix}\lambda_k\\ \lambda_v\end{bmatrix} (C \begin{bmatrix}U_k\\ U_v\end{bmatrix})$$

- Calculating λ_k and λ_v

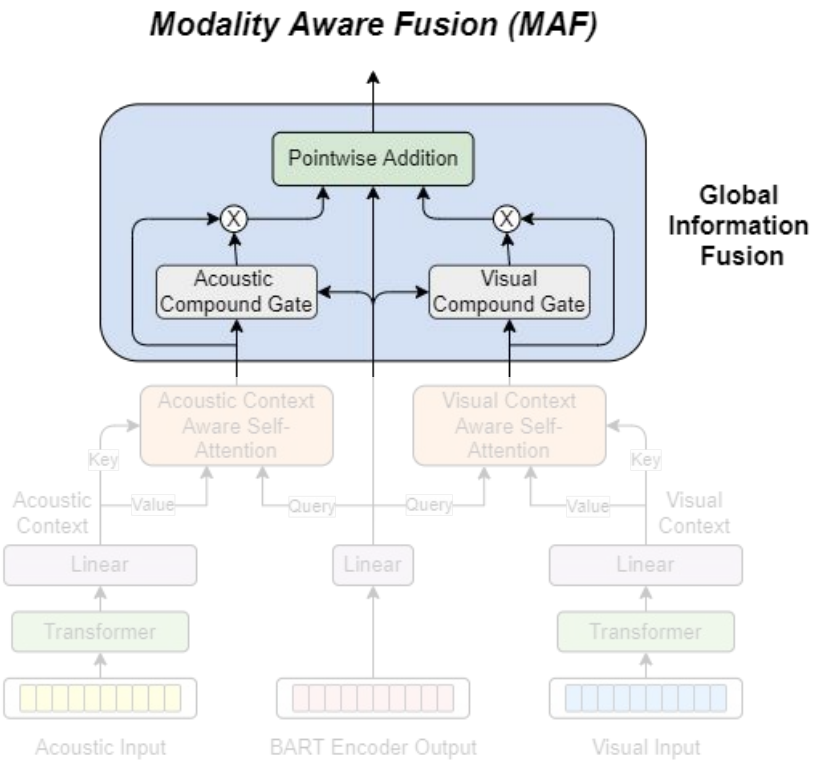
$$\begin{bmatrix}\lambda_k\\ \lambda_v\end{bmatrix} = \sigma(\begin{bmatrix}K\\ V\end{bmatrix} \begin{bmatrix}W_{k_1}\\ W_{v_1}\end{bmatrix} + C \begin{bmatrix}U_k\\ U_v\end{bmatrix} \begin{bmatrix}W_{k_2}\\ W_{v_2}\end{bmatrix})$$

- Information infused attention

$$H_a = Softmax(\frac{Q\hat{K}_a^T}{\sqrt{d_k}})\hat{V}_a$$

$$H_v = Softmax(\frac{Q\hat{K}_v^T}{\sqrt{d_k}})\hat{V}_v$$

Proposed Architecture: MAF-TAV



- Combining acoustic and visual modalities

$$g_a = [H \oplus H_a]W_a + b_a$$
$$g_v = [H \oplus H_v]W_v + b_v$$

- Final multimodal information

$$\hat{H} = H + g_a \odot H_a + g_v \odot H_v$$

Results

Mode	Model	R1	R2	RL	B1	B2	B3	B4	M	BS
Textual	**RNN**	29.22	7.85	27.59	22.06	8.22	4.76	2.88	18.45	73.24
	Transformers	29.17	6.35	27.97	17.79	5.63	2.61	0.88	15.65	72.21
	PGN	23.37	4.83	17.46	17.32	6.68	1.58	0.52	23.54	71.90
	mBART	33.66	11.02	31.50	22.92	10.56	6.07	3.39	21.03	73.83
	BART	36.88	11.91	33.49	27.44	12.23	5.96	2.89	26.65	76.03
Multimodality	**MAF-TA$_M$**	39.02	15.90	36.83	31.26	16.94	11.54	7.72	29.05	77.06
	MAF-TV$_M$	39.47	16.78	**37.38**	32.44	17.91	12.02	7.36	29.74	77.47
	MAF-TAV$_M$	38.52	14.13	36.60	30.50	15.20	9.78	5.74	27.42	76.70
	MAF-TA$_B$	38.21	14.53	35.97	30.58	15.36	9.63	5.96	27.71	77.08
	MAF-TV$_B$	37.48	15.38	35.64	30.28	16.89	10.33	6.55	28.24	76.95
	MAF-TAV$_B$	**39.69**	**17.10**	37.37	**33.20**	**18.69**	**12.37**	**8.58**	**30.40**	**77.67**

Experimental results for MAF-TAV

Model	R1	R2	RL	B1	B2	B3	B4	M	BS
MAF-TAV$_M$	**38.52**	14.13	36.60	30.50	15.20	9.78	5.74	27.42	76.70
- MCA2 + CONCAT1	37.56	**14.85**	34.90	30.16	15.76	**10.12**	**6.82**	**28.59**	76.59
- MAF + CONCAT2	17.22	1.70	14.12	13.11	2.11	0.00	0.00	9.34	66.64
- MCA2 + DPA	36.43	13.04	33.75	28.73	14.02	8.00	4.89	25.60	75.58
- GIF	36.37	13.85	34.92	28.49	14.34	9.00	6.16	25.75	**76.86**
MAF-TAV$_B$	**39.69**	**17.10**	**37.37**	**33.20**	**18.69**	**12.37**	**8.58**	**30.40**	**77.67**
- MCA2 + CONCAT1	36.88	13.21	34.39	29.63	14.56	8.43	4.84	26.15	76.08
- MAF + CONCAT2	21.11	2.31	19.68	12.44	2.44	0.73	0.31	9.51	69.54
- MCA2 + DPA	38.84	14.76	36.96	30.23	15.95	9.88	5.83	28.04	77.20
- GIF	39.45	14.85	37.18	31.85	15.97	9.62	5.47	28.87	77.54

Ablation results on MAF-TAV

Human Evaluation

- **Coherence:** Measures how well the explanations are organized and structured.
- **Related to dialogue:** Measures whether the generated explanation adheres to the topic of the dialogue.
- **Related to sarcasm:** Measures whether the explanation is talking about something related to the sarcasm present in the dialogue.

	Coherency	Related to dialogue	Related to sarcasm
mBART	2.57	2.66	2.15
BART	2.73	2.56	2.18
MAF-TA$_B$	2.95	2.91	2.51
MAF-TV$_B$	3.01	**3.11**	2.66
MAF-TAV$_B$	**3.03**	**3.11**	**2.77**

Qualitative Results

ROSESH: What nonsense? Mujhe kon hunter marega? *What nonsense? Who will beat me with a hunter?*
INDRAVARDHAN: Me, me marunga. Kyunki 51 saal baad Maya to mar chuki hogi. Me kisi tarah zinda reh lunga or kahunga ki 'le mar jhadhu, mar, mar jhadhu, mar mar' *I will beat you. Because after 51 years, Maya would be dead. I'll somehow survive and say 'sweep here, sweep here, sweep'*

Gold	Maya Monisha ko tana marti hai safai ka dhyan na rakhne ke liye *Maya taunts Monisha for not keeping a check of cleanliness*
BART	Maya Monisha ko tumaari burayi nahi karta. *Maya doesn't blame you for Monisha*
MAF-TAV$_B$	Maya implies ki Monisha bohot ghar mein bahar nahi kar sakati. *Maya implies that Monisha very in home cannot do outside.*

Incoherent explanation

SAHIL: Ab tumne ghar ki itni saaf safai ki hai and secondly us Karan Verma ke liye pasta, lasagne, caramel custard banaya. *Now you have cleaned the house so much and secondly made pasta, lasagne, caramel custard for that reason Verma.*
MONISHA: Walnut brownie bhi. *And walnut brownie too.*
SAHIL: Walnut brownie, matlab wo khane wali? *You mean edible walnut brownie?*

Gold	Sahil monisha ki cooking ka mazak udata hai *Sahil makes fun of Monisha's cooking.*
BART	Monisha sahil ko walnut brownie ki matlab wo khane wali. *Walnut Brownie to Monisha Sahil means she eats*
MAF-TAV$_B$	Sahil monisha ki cooking ka mazak udata hai *Sahil makes fun of Monisha's cooking.*

Explanation related to dialogue

MONISHA: Ladki ka naam Ajanta Kyon Rakha? *Why did they named the girl Ajanta?*
INDRAVARDHAN: Kyunki uski maa ajanta caves dekh rahi thi Jab vo Paida Hui haha. *Because her mother must be watching the Ajanta caves when she was born haha.*

Gold	Indravadan Ajanta ke naam ka mazak udata hai *Indravardhan makes fun of Ajanta's name*
BART	Indravardhan Monisha ko taunt maarta hai ki uski maa ajanta caves dekh rahi thi Jab vo Paida Hui *Indravardhan taunts Monisha as her mother was watching Ajanta Caves when she was born.*
MAF-TAV$_B$	Indravadan ajanta ke naam ka mazak udata hai *Indravardhan makes fun of Ajanta's name*

Explanation related to sarcasm

Affect Analysis

Question: Can explanation help in the comprehension of various affects?

Three tasks in question:

- Sarcasm detection
- Humour identification
- Emotion recognition

	WITS	sWITS		hWITS		eWITS			
	#S	**#NS**	**#S**	**#NH**	**#H**	**#Neutral**	**#Sadness**	**#Joy**	**#Anger**
Train	1792	1669	1792	2795	995	1590	1147	623	429
Val	224	213	224	362	112	196	133	87	57
Test	224	218	224	367	106	195	141	70	67
Total	2240	2100	2240	3524	1213	1981	1421	780	553

Affect Analysis- Results

Model	Use of Explanation		Sarcasm				Humor				Emotion		
	Train	Test	P	R	F1	Acc	P	R	F1	Acc	P	R	F1
None	**0**	**0**	0.57	0.68	0.62	0.57	0.69	0.78	**0.73**	**0.87**	**0.8**	0.78	0.78
MAF	**1**	**0**	0.58	0.73	0.65	0.6	0.57	**0.87**	0.69	0.81	0.78	0.78	0.78
	1	**1**	**0.66**	**0.77**	**0.71**	**0.68**	**0.73**	0.71	0.72	**0.87**	0.78	**0.81**	**0.79**

Affect Analysis- Results

Model	Use of Explanation		Sarcasm				Humor				Emotion		
	Train	Test	P	R	F1	Acc	P	R	F1	Acc	P	R	F1
None	**0**	**0**	0.57	0.68	0.62	0.57	0.69	0.78	0.73	0.87	0.8	0.78	0.78
MOSES	**1**	**0**	0.65	0.71	0.68	0.66	**0.84**	0.63	0.72	**0.89**	0.79	0.78	0.78
	1	**1**	**0.70**	**0.83**	**0.76**	**0.73**	0.72	0.77	**0.75**	0.88	**0.81**	0.80	**0.80**

Explaining Stereotypes in HateSpeeches

Explaining Toxic Text via Knowledge Enhanced Text Generation, NAACL-2022

what's the most perfect thing? a rainbow, because it has no black on it.

Implied stereotype

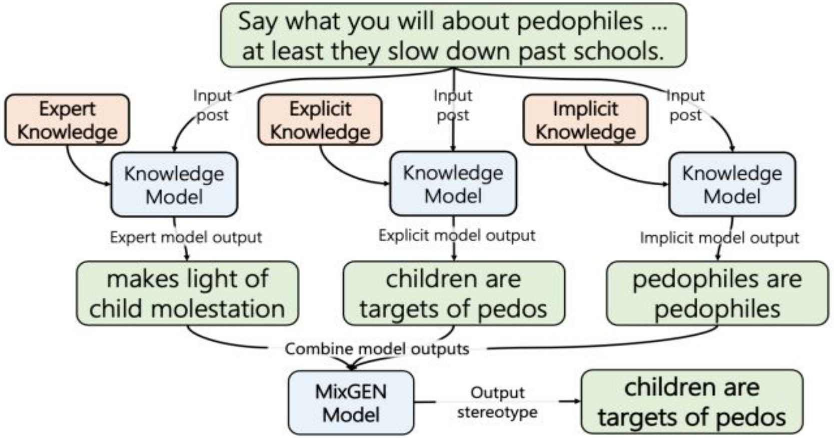
black people are worthless; black people are inferior;

what's can a white man say that a black man can't? "thanks officer!" and "hey dad!"

Implied stereotype

black folks are criminals; black fathers are not present in children's lives; black men abandon their kids;

Explaining Toxic Text via Knowledge Enhanced Text Generation, NAACL-2022

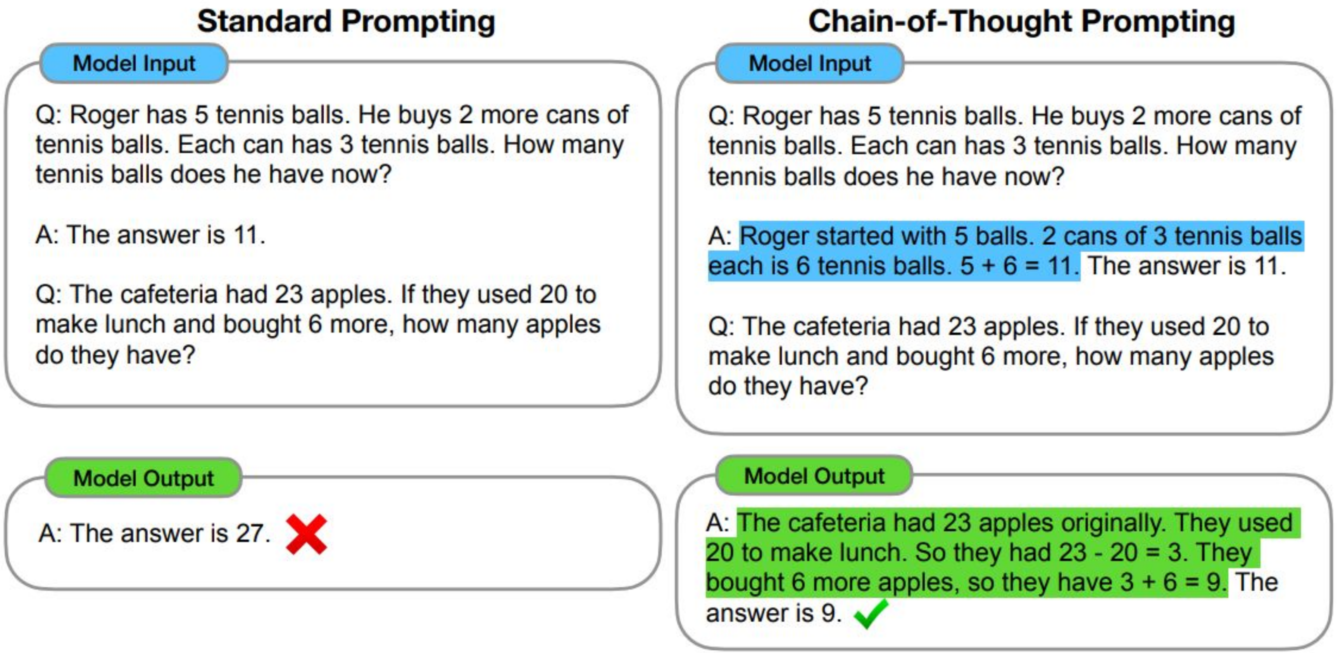


BART

Type	Description	Pros	Cons
Expert Knowledge	Expert judgements from annotations on the input.	• Easy-to-use. • High quality and accurate knowledge.	• Sparse and hard to obtain. • Difficult to get a lot of diverse knowledge.
Explicit Knowledge	External knowledge sourced from knowledge bases and graphs. Restructured, determinstic interpretations of raw text sources.	• Many knowledge bases and graphs exist. • Easy to query due to symbolic representation. • Explainable since knowledge source is known.	• Fixed knowledge, thus limited retrieval diversity. • Explicitly constructed, thus may not be complete.
Implicit Knowledge	External knowledge sourced from large pretrained models. Probabilistically generated from raw text sources	• Easy to retrieval. • Probabilistic, thus increasing diversity in retrieval.	• Low explainability. • Low quality, since the knowledge is implicitly learned.

Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, NIPS-2022

- Chain of thought — a series of intermediate reasoning steps for an output.
  - significantly improves the ability of large language models to perform complex reasoning



Math Word Problems (free response)

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

Math Word Problems (multiple choice)

Q: How many keystrokes are needed to type the numbers from 1 to 500?
Answer Choices: (a) 1156 (b) 1392 (c) 1480 (d) 1562 (e) 1788

A: There are 9 one-digit numbers from 1 to 9. There are 90 two-digit numbers from 10 to 99. There are 401 three-digit numbers from 100 to 500. 9 + 90(2) + 401(3) = 1392. The answer is (b).

CSQA (commonsense)

Q: Sammy wanted to go to where the people were. Where might he go?
Options: (a) race track (b) populated areas (c) desert (d) apartment (e) roadblock

A: The answer must be a place with a lot of people. Race tracks, desert, apartments, and roadblocks don't have a lot of people, but populated areas do. So the answer is (b).

StrategyQA

Q: Yes or no: Would a pear sink in water?

A: The density of a pear is about 0.6 g/cm^3, which is less than water. Thus, a pear would float. So the answer is no.

Date Understanding

Q: The concert was scheduled to be on 06/01/1943, but was delayed by one day to today. What is the date 10 days ago in MM/DD/YYYY?

A: One day after 06/01/1943 is 06/02/1943, so today is 06/02/1943. 10 days before today is 05/23/1943. So the answer is 05/23/1943.

Sports Understanding

Q: Is the following sentence plausible? "Joao Moutinho caught the screen pass in the NFC championship."

A: Joao Moutinho is a soccer player. The NFC championship is part of American football, not soccer. So the answer is no.

SayCan (Instructing a robot)

Human: How would you bring me something that isn't a fruit?

Explanation: the user wants something to eat that isn't a fruit. An energy bar is not a fruit, so I will bring the user an energy bar.
Plan: 1. find(energy bar) 2. pick(energy bar) 3. find(user) 4. put(energy bar) 5. done().

Last Letter Concatenation

Q: Take the last letters of the words in "Lady Gaga" and concatenate them.

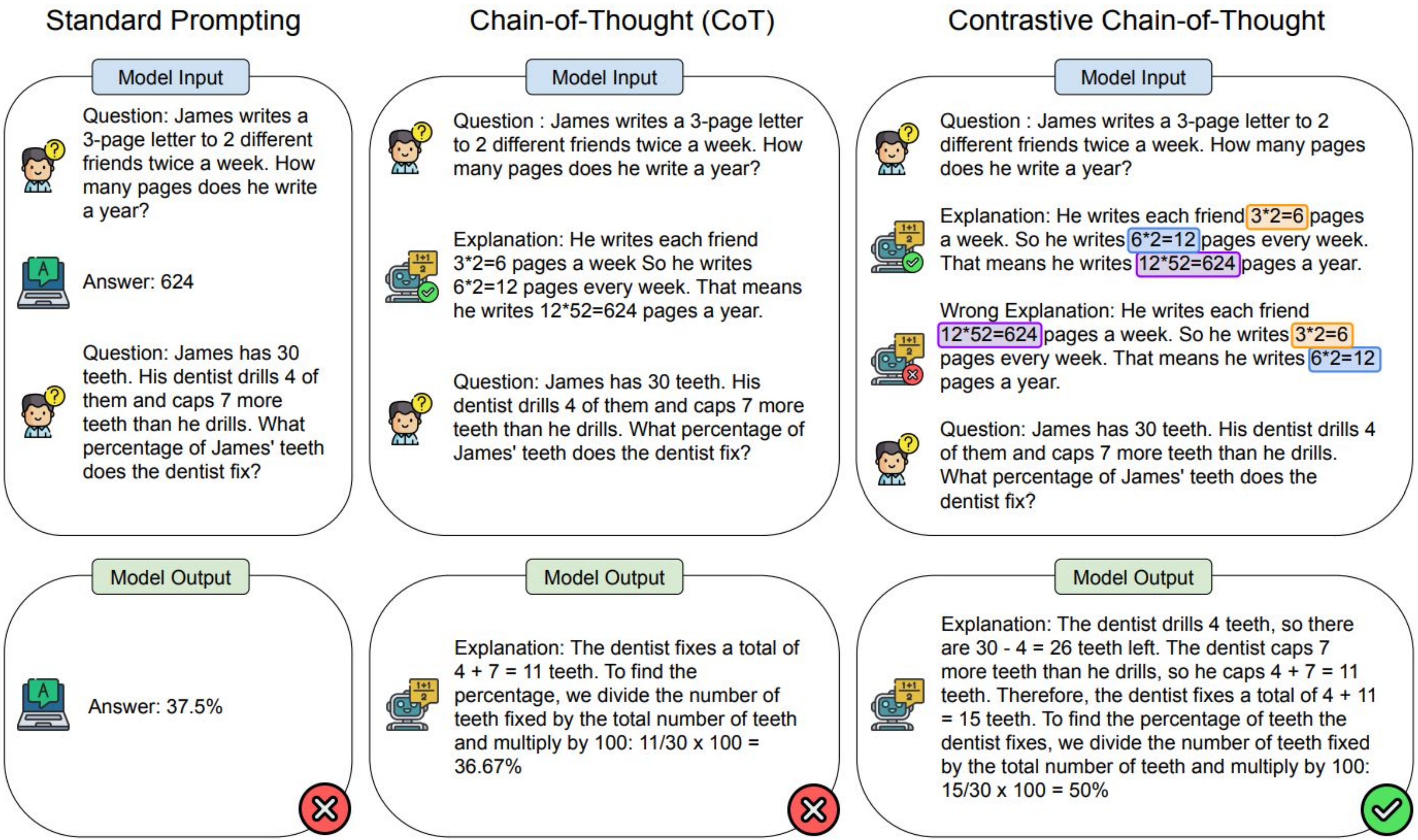
A: The last letter of "Lady" is "y". The last letter of "Gaga" is "a". Concatenating them is "ya". So the answer is ya.

Coin Flip (state tracking)

Q: A coin is heads up. Maybelle flips the coin. Shalonda does not flip the coin. Is the coin still heads up?

A: The coin was flipped by Maybelle. So the coin was flipped 1 time, which is an odd number. The coin started heads up, so after an odd number of flips, it will be tails up. So the answer is no.

Contrastive Chain-of-Thought Prompting



Using ChatGPT to generate contextual toxic content

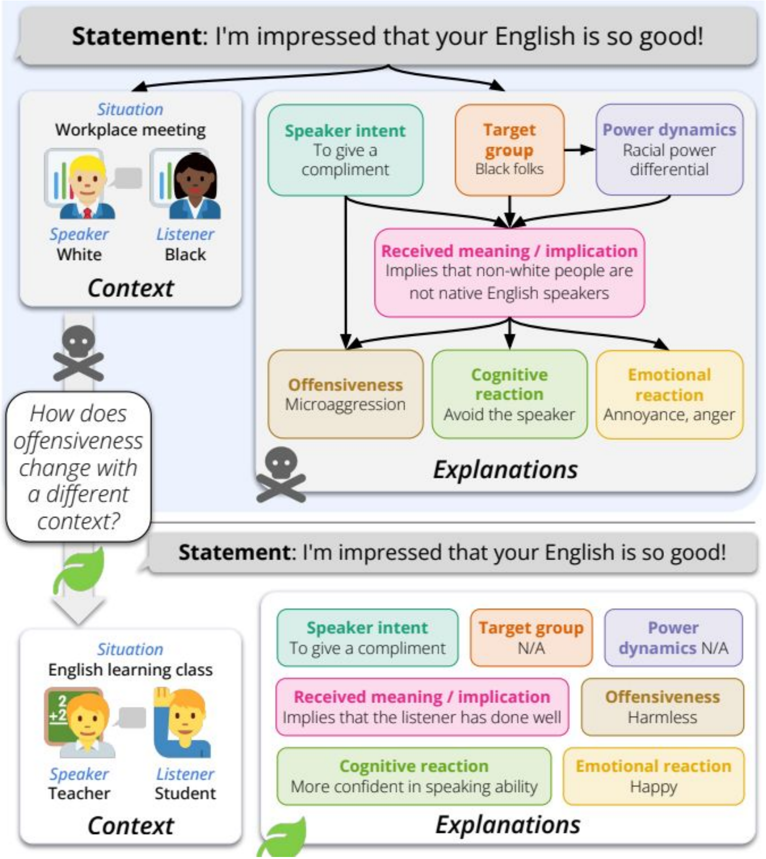


Fig 1: Different output for toxicity based on context [1]

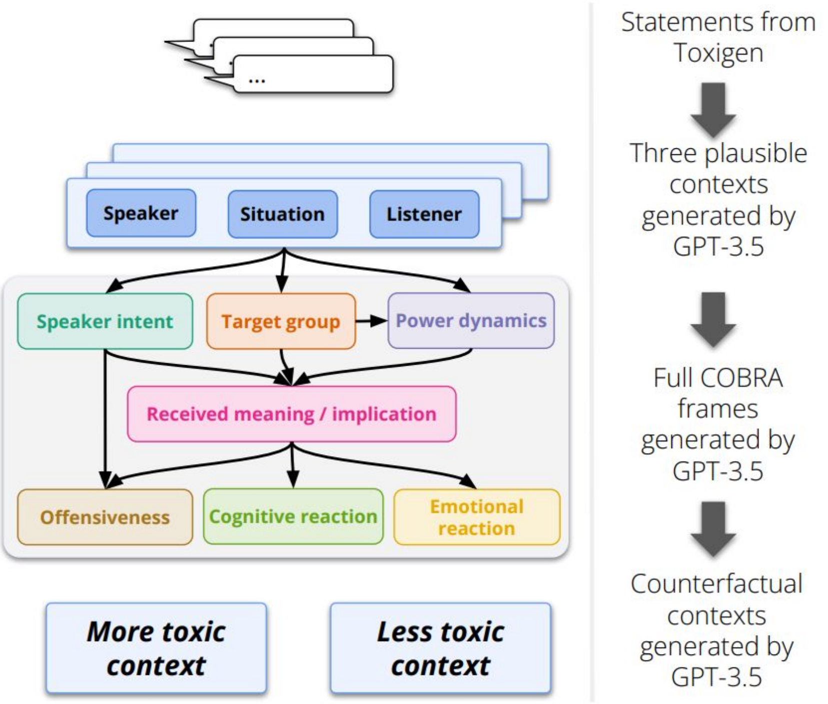


Fig 2: Using ChatGPT to generate large scale scenario-based toxicity[1]

[1] Large Scale Crowdsourcing and Characterization of Twitter Abusive Behavior

Thank You!